

Digital Services

Powered by InSite 2.0



The digital revolution in healthcare has arrived. You experience it every day – with technologies that allow you to diagnose, monitor and treat patients more accurately and quickly than ever before.

Digital Service

GE Healthcare has some of the world's most advanced portfolios of intelligent service and asset management tools for healthcare providers. We will take your organization to new levels of quality and efficiency, enabling you to compete in your marketplace in ways you never dreamed possible.

On-demand Digital Service

It's all about speed. When you contact GE for service, our online engineers can electronically link to your system and work to quickly get you back up and running. If a Field Engineer needs to be dispatched, he or she will be prepared with a system diagnosis – and if required, have the right parts on hand. In addition, InSite 2.0 customers with a PriorityPass agreement will have direct access to technical support and can bypass the customer care center and phone menu for one of the fastest technical responses in the industry.

Proactive Digital Service

GE can proactively monitor key operational parameters of your medical systems to provide early warning of potential issues and head off costly, unscheduled downtime. Our online engineers have the ability to recalibrate key operational parameters to help ensure optimal system performance, or they can dispatch a field engineer to assist in mitigating the issue.

Additionally, automated software downloads ensure you will receive software updates and upgrades in a more timely manner to keep your medical systems working efficiently. Software downloads also significantly reduce the time it takes to upgrade your GE Healthcare devices, which means scheduled system downtime and clinical workflow interruptions are greatly reduced.

Frequently asked questions about InSite 2.0

How does GE Healthcare connect to my equipment?

The two major technical components of InSite 2.0 are the Agent and the Server. The Agent is installed on GE Healthcare equipment at customer sites while the Server resides within GE Healthcare.

The Agent

A piece of embedded software that handles InSite 2.0 on the GE Healthcare equipment. The Agent is very small. Its role is to: monitor device performance data on an ongoing basis, establish secure communications to the Server via the Internet, and send fault information and log files to the Server.

The Server

The Server uses secure Web Services to communicate with the Agent.

It processes the performance and fault information provided by the Agent. It is the management console for InSite 2.0 that allows our support team to: consolidate and analyze data; run diagnostics remotely; set automated rules for taking action when certain events occur; set up software updates for distribution; and specify security and access to customer devices.

How does an InSite 2.0 installation work?

The Agent establishes connectivity from behind the safety of your corporate firewall, adhering to all security policies set up by your network administrators. To your network, the Agent is just another computer on the LAN. To set up the InSite 2.0 Agent at your site, the only networking requirements are as follows:

- A physical connection or a route to an existing enterprise LAN
- Allow outbound Internet access for the device using HTTPS protocol

A GE Healthcare Field Engineer will install a product or software upgrade that incorporates the Agent and configures network connections according to your IT requirements.

What type of Internet connections will InSite 2.0 support?

InSite 2.0 utilizes your existing outbound broadband Internet connection. It uses Secure Sockets Layer (SSL) and complies with your existing firewall rules and Web proxies. The product can use either a static or dynamic IP address, and no VPN configuration is necessary. However, InSite 2.0 can run over an existing VPN connection if VPN is required by your facility.

Who is connecting to my GE Healthcare equipment?

There are four types of engineers who deliver digital service. All connections are initiated by the customer's device and are logged regardless of who is connecting or why.

- **OnLine Engineer (OLE):** Technical expert who remotely supports the equipment. Thanks to digital tools, OLEs can analyze problems with the machines, isolate defective parts, and in some cases remotely fix the equipment.
- **OnLine Application Engineer (OLA):** Application specialist who remotely supports the customer. Thanks to digital tools, OLAs can better understand customer problems.
- **eFE:** After completing a comprehensive eFE training program, your Field Engineer can gain access to the digital service tools that enable his/her access to your equipment remotely. With these tools, your Field Engineer can often diagnose an equipment issue before arriving on-site with the appropriate preventative or corrective action.
- **Product Engineering:** GE Healthcare Product Engineers may access your equipment to gather system performance data to better understand capabilities of GE Healthcare equipment installed at and used by our GE Healthcare customers.

Can all my GE Healthcare equipment come online with InSite 2.0?

InSite 2.0 represents the next wave in remote serviceability. We are working to incorporate InSite 2.0 into our portfolio of medical systems. Contact your GE Healthcare account manager for current compatibility information.

Will InSite 2.0 impact my network performance?

The Agent polls the Server at regular intervals (minutes or hours – a very long interval in machine time). It can be set up to transmit data to the Server only at scheduled times or when an error code indicates that immediate notification is needed. The messaging protocol is designed to deliver small, efficient (0.5-3Kbytes) messages to minimize bandwidth utilization. This means that InSite 2.0 communications will not significantly impact network bandwidth.

Are InSite 2.0 Digital Services secure?

The goal of InSite 2.0 is to support existing network standards and security practices while maintaining a high level of security for digital service transmissions. In general, no special configurations are necessary to accommodate InSite 2.0 or compromise your firewall in any way.

This is possible through a three-layer security architecture based on Web services that achieves both transparency and security. InSite 2.0 employs security at the device, network and enterprise layers, built using technology specifically designed for secure, efficient device management communications. This includes a hardened software design for application and data security with support for widely used industry standards, such as TCP/IP, HTTPS, SOAP, and XML. Browser interfaces are also based on industry standard Java and HTML, while the enterprise software is built using J2EE.

The key security features of InSite 2.0 connectivity are as follows:

- **No special port required**

Unlike some remote access products, In-Site 2.0 does not mandate the use of a specific port number. This is commonly referred to as “poking a hole in the firewall,” and requires special, non-standard configuration by network administrators.

- **No fixed IP address required**

The Agent supports DHCP, so there is no need to assign fixed IP addresses to devices. Typical firewall configurations allow for outbound connections to be initiated securely

from “behind” the firewall, over port 443 (HTTPS). The Agent communicates through the firewall using port 443 and initiates all communication to the GE Healthcare InSite 2.0 Server. Therefore, equipment can communicate to the Server without revealing its true IP address, in the same way that a Web browser accesses a Web site. Since all such communication is outbound, your corporate network does not need to accept any connections from the outside or open up ports in order to allow InSite 2.0 to do its job. Equipment that is InSite 2.0-enabled simply will not accept connections from any system outside the corporate firewall.

- **The Agent always connects to a known IP address (the Server)**

An important issue for IT managers is whether the connection is with a trusted party or with an unknown entity. If the Agent had to listen on a TCP port while waiting for a connection, this would make the device a potential target for unauthorized access. With InSite 2.0, the ability to identify the Server is guaranteed because the Server is visible to the Agent only via a known IP address, and therefore the Agent never has to “listen” on a port. The Agent communicates only through the secure tunnel it has established to the named and trusted InSite 2.0 Server. Therefore, identity is not an issue.

- **The Agent is flexible**

All of the features described above contribute to flexibility and compatibility in accommodating changing network infrastructures. The Agent is not dependent on a static IP address or subnets and supports corporate network infrastructures that require Internet proxy servers.

- **Only InSite 2.0 users and applications can use the tunnel**

Once the Agent has established a secure tunnel, the connection is visible only to InSite 2.0 clients and services (applications or users). This means that unauthorized clients and services cannot use the connection even if they manage to “see” it.

- **Security that is less costly and more convenient than VPN**

Since the Agent is responsible for initiating two-way communication in a manner compliant with the secure

computing environment at the customer site, there is no need for a Virtual Private Network (VPN). The only requirement is an Internet connection. This is a far less complicated and less costly approach than having to supply, configure and maintain VPN hardware. However, InSite 2.0 can run over an existing VPN connection if VPN is required by your facility.

- **Secure data transmission**

The Agent communicates with the Server via transmissions that require password authentication to validate the identity of devices exchanging information with the enterprise. All data transmissions are encrypted using 128-bit or higher (less where government restrictions apply) Secure Socket Layer (SSL) protocol.

- **Secure data collection**

The Agent only collects device data relevant to the performance of your equipment.

- **Reducing security vulnerabilities**

InSite 2.0 connectivity can enable better protection through automated software management. Automated software management is available on certain InSite 2.0 capable equipment. With this automated capability, the GE Healthcare support team can ensure that urgent software patches are applied as soon as possible after validation to reduce the amount of time that your operations might be vulnerable to attack. On occasions such as this, when GE Healthcare is updating your equipment, you can be assured that the patches are coming from the InSite 2.0 Server, which is dedicated to supporting only GE Healthcare customers.

- **Secure server access**

At the enterprise level, InSite 2.0 allows only authorized users at GE Healthcare to log in with username and password authentication, via single sign-on. This ensures only qualified and trained engineers are working on your system. In addition, all user and system interactions are logged.

What about patient safety and privacy?

Patient safety

GE Healthcare Digital Services delivery operations do not compromise patient safety. GE Healthcare has Digital Service delivery policies and procedures in place to help protect patient safety.

Patient privacy

In some cases, GE Healthcare may encounter protected health information (PHI) as part of our troubleshooting procedures or under data access rights granted to GE Healthcare. Access to this data is limited to GE Healthcare authorized personnel only. GE complies with all patient privacy regulations as received by applicable government regulations.

What about security issues on my GE Healthcare equipment?

GE Healthcare publishes up-to-date listings of product operating system vulnerabilities, virus protection issues, and network security information at http://apps.gehealthcare.com/product_security/

Visit this site for current information on product-side security issues.

The goal of InSite 2.0 is to support existing network standards and security practices while maintaining a high level of security for digital service transmissions. In general, no special configurations are necessary to accommodate InSite 2.0 or compromise your firewall in any way.

GE InSite 2.0 Digital Services*– delivering value across your organization

Hospital Administration

Staff Productivity

- Digital software downloads can reduce time to upgrade by up to 93%
- Remote diagnostics and assistance reduces time to repair by up to 70%
- iLinq™ and PriorityPass (for InSite 2.0 Digital Services) give your employees priority access to and response from technical support engineers, reducing wait times by as much as 84%

Equipment Uptime

- Improved equipment uptime when using digital software updates and system event notification
- JCAHO/HIPAA support with improved auditing capabilities of connection activities

Predictable Costs

- InSite™ 2.0 Connectivity and Digital Remote Assist Fix is included with products under warranty or contract

Information Technologies

Security

- Secure Connectivity – 128bit SSL encryption
- Improved access to new software via digital downloads
- No inbound holes in firewall
- Control over remote access
- Improved audit reporting of connections

Access to Latest Released Patches

- Improved access to the latest security and software patches via digital downloads

Cost Reduction

- Less maintenance \$/hr than VPN

Bio-Medical Engineering

Clinical Excellence

- Improved access to new software via digital downloads
- Reduced impact to clinical workflow when updates are performed

Staff Productivity

- System Alerts to assist in diagnosing problems, which translates into reduced time to repair
- Digital software downloads can reduce time to upgrade by up to 93%
- iLinq and PriorityPass (for InSite 2.0 Digital Services) give your employees priority access to and response from technical support engineers, reducing wait times by as much as 84%
- Remote diagnostics and assistance reduces time to repair by up to 70%

Clinical Staff

Equipment Confidence

- 24/7 monitoring of equipment vital signs
- Equipment working when you need it
- Improved access to new software via digital downloads
- Reduced impact to workflow when updates are performed

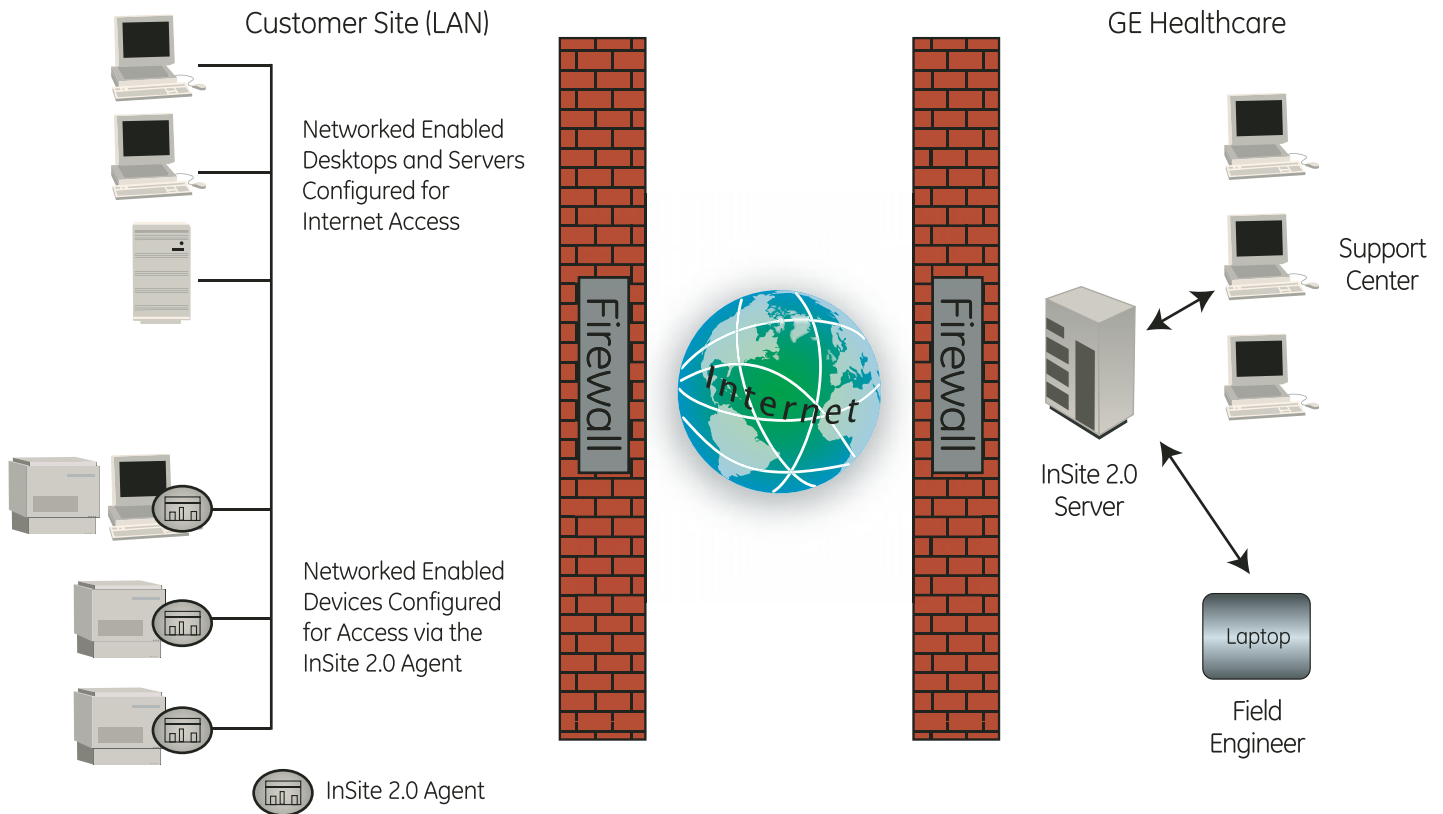
Clinical Assistance

- On Product assistance for product “How do I...” questions when using clinical iLinq

**InSite 2.0 feature availability varies between products and Service Support Agreement types.*

InSite™ 2.0 links your critical medical systems to GE Healthcare's Support Center via a secure Internet connection to enable the on-demand and proactive Digital Services that help you achieve your clinical and business objectives.

Here's how it works



InSite 2.0 is comprised of both an Agent that resides on your GE Healthcare system and a Server within GE Healthcare. The Agent on the GE Healthcare system initiates contact with the GE Healthcare Server at established time intervals to transfer key operational parameters.

Key InSite 2.0 security features

- No additional firewall ports required
- Less expensive and easier to install than a typical Virtual Private Network (VPN)
- 128 bit Secure Sockets Layer (SSL) data encryption and multi-level authentication, permission and role definitions
- Web services standard protocol using only HTTPS Port 443
- Connection is initiated from customer end, giving customer full access control

- Only GE authorized users, under stringent monitoring and tracking, can access customer system and only to collect relevant equipment performance data for analysis

Digital Service is the next generation of intelligent GE service and asset management technologies – as advanced as the systems they support – that will enable your enterprise to deliver higher quality, more cost-effective healthcare. What could be smarter than that?

©2006 General Electric Company – All rights reserved.

General Electric Company reserves the right to make changes in specifications and features shown herein, or discontinue the product described at any time without notice or obligation.

GE, GE Monogram and InSite™ are trademarks of General Electric Company.

GE Medical Systems Information Technologies, a General Electric company, going to market as GE Healthcare.

For more than 100 years, healthcare providers worldwide have relied on GE Healthcare for medical technology, services, and productivity solutions. So no matter what challenges your healthcare system faces, you can always count on GE to help you deliver the highest quality healthcare. For details, please contact your GE representative today.

GE Healthcare
3000 North Grandview
Waukesha, WI 53188
U.S.A.

www.gehealthcare.com



imagination at work